

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

**Product Name** 1 ppm Ammonia Standard

**Product No** 2210001

**Pure substance/mixture** Mixture

### Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Use as laboratory reagent

**Uses advised against** No Information available

### **Supplier**

|                                                                                                                                                                                                    |                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Fisher Scientific Korea<br>D5,D6, Incheon Airport Logistics<br>Complex<br>150, Gonghangdong-Ro Mapo-Gu 296<br>Beon-Gil<br>Jung-Gu, Incheon 400-340<br>Tel: +82-2-6196-5500<br>Fax: +82-2-6196-5501 | Thermo Fisher Scientific©<br>Water and Lab Products<br>22 Alpha Road<br>Chelmsford, MA 01824, USA<br>1-978-232-6000 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

### **E-mail address**

Chem.KR@thermofisher.com info.water@thermo.com

### Emergency Telephone Number

**Emergency telephone:** Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

2210001

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

Based on available data, the classification criteria are not met

## **Environmental hazards**

Based on available data, the classification criteria are not met

## **Label Elements**

## **Other Hazards**

## **NFPA**

Health  
0

Flammability  
0

Instability  
0

Physical hazards  
N/A

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### **3.2. Mixtures**

| Component         | Common Name              | CAS-No     | Index No. | Weight % |
|-------------------|--------------------------|------------|-----------|----------|
| Water             | No information available | 7732-18-5  | KE-35400  | 99 - 100 |
| Ammonium Chloride | No information available | 12125-02-9 | KE-01645  | <0.1     |

## **SECTION 4: FIRST AID MEASURES**

### **First aid measures**

#### **General Advice**

Use first aid treatment according to the nature of the injury. Get medical attention immediately if symptoms occur. Show this safety data sheet to the doctor in attendance.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### **Inhalation**

Move to fresh air. Get medical attention immediately if symptoms occur.

#### **Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### **Self-Protection of the First Aider**

No special precautions required.

### **Most important symptoms and effects, both acute and delayed**

#### **Most important symptoms and effects**

None reasonably foreseeable

### **Indication of any immediate medical attention and special treatment needed**

#### **Notes to Physician**

Treat symptomatically

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

## SECTION 5: FIREFIGHTING MEASURES

### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### Unsuitable Extinguishing Media

No information available

### Specific Hazards Arising from the Chemical

No information available.

### Explosion Data

**Sensitivity to Mechanical Impact** None

**Sensitivity to Static Discharge** None

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment.

#### Environmental Precautions

Should not be released into the environment. See Section 12 for additional ecological information.

### Methods and Material for Containment and Cleaning Up

#### Methods for Containment

Prevent further leakage or spillage if safe to do so.

#### Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

#### Handling

Wear personal protective equipment  
Ensure adequate ventilation  
Avoid contact with skin, eyes and clothing  
Avoid ingestion and inhalation

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage

Keep container tightly closed in a dry and well-ventilated place  
Store at room temperature in the original container  
Keep away from direct sunlight

#### Incompatible Products

No information available

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component         | CAS-No     | Korea                                                   | ACGIH TLV                                               | OSHA PEL                                                                    |
|-------------------|------------|---------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------|
| Water             | 7732-18-5  | Not listed                                              | Not listed                                              | Not listed                                                                  |
| Ammonium Chloride | 12125-02-9 | STEL: 20 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) STEL: 20 mg/m <sup>3</sup> |

| Component | CAS-No | European Union | The United Kingdom | Germany |
|-----------|--------|----------------|--------------------|---------|
|-----------|--------|----------------|--------------------|---------|

2210001

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

|                   |            |            |                                                                     |            |
|-------------------|------------|------------|---------------------------------------------------------------------|------------|
| Water             | 7732-18-5  | Not listed | Not listed                                                          | Not listed |
| Ammonium Chloride | 12125-02-9 | Not listed | STEL: 20 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr | Not listed |

## Biological Exposure Indices

| Component         | CAS-No     | Biological Exposure Indices |
|-------------------|------------|-----------------------------|
| Water             | 7732-18-5  | Not listed                  |
| Ammonium Chloride | 12125-02-9 | Not listed                  |

## Appropriate engineering controls

**Engineering Measures** None under normal use conditions

## Individual protection measures, such as personal protective equipment

**Personal protective equipment** Use only those certified by the Korea Occupational Safety and Health Administration.

**Eye/face Protection** Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:  
Face-shield.

**Skin and Body Protection** Wear protective gloves/clothing.

**Respiratory Protection** None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical State | Liquid                   |
| Appearance     | Clear                    |
| Odor           | None                     |
| Odor Threshold | No information available |
| pH             | 6.85                     |
| PH Range       | No information available |

| Property                     | Values                   | Remarks • Method |
|------------------------------|--------------------------|------------------|
| Melting point/freezing point | No information available |                  |
| Boiling Point/Range          | No information available |                  |
| Flash Point (High in °C)     | N/A                      |                  |
| Evaporation Rate             | No information available |                  |
| Flammability (solid, gas)    | No information available |                  |
| Flammability Limit in Air    |                          |                  |
| Upper flammability limit:    | No information available |                  |
| Lower flammability limit:    | No information available |                  |
| Vapor pressure               | No information available |                  |
| Vapor Density                | No information available |                  |
| Specific Gravity             | No information available |                  |
| Water Solubility             | Soluble in water         |                  |
| Solubility in other solvents | No information available |                  |
| Partition coefficient        | No information available |                  |
| Autoignition Temperature     | -                        |                  |
| Decomposition Temperature    | No information available |                  |
| Kinematic viscosity          | No information available |                  |
| Dynamic viscosity            | No information available |                  |

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

|                             |                          |
|-----------------------------|--------------------------|
| <b>Explosive Properties</b> | No information available |
| <b>Oxidizing Properties</b> | No information available |

## Other Information

|                         |                          |
|-------------------------|--------------------------|
| <b>Softening Point</b>  | No information available |
| <b>Molecular Weight</b> | No information available |
| <b>VOC Content(%)</b>   | No information available |
| <b>Density</b>          | No Information available |
| <b>Bulk Density</b>     | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

No Information available

### Chemical Stability

Stable under normal conditions

### Possibility of Hazardous Reactions

None under normal processing

### Conditions to Avoid

Extremes of temperature and direct sunlight.

### Incompatible Materials

No information available

### Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 11: TOXICOLOGICAL INFORMATION

### **Product Information**

#### Information on likely routes of exposure

|                   |                                                |
|-------------------|------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.             |
| <b>Ingestion</b>  | No known effect based on information supplied. |
| <b>Eyes</b>       | Not an expected route of exposure.             |
| <b>Skin</b>       | No known effect based on information supplied. |

#### Information on Toxicological Effects

|                 |                          |
|-----------------|--------------------------|
| <b>Symptoms</b> | No information available |
|-----------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                 |                           |
|---------------------------------|---------------------------|
| <b>Sensitization</b>            | No information available  |
| <b>Mutagenic Effects</b>        | No information available  |
| <b>Carcinogenicity</b>          | No information available. |
| <b>Reproductive Effects</b>     | No information available  |
| <b>STOT - single exposure</b>   | No information available  |
| <b>STOT - repeated exposure</b> | No information available  |

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

**Target Organs** None known.

**Aspiration hazard** No information available

## Numerical measures of toxicity - Product Information

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity

0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

| Component                       | Freshwater Algae | Freshwater Fish                                                                               | Water Flea                            |
|---------------------------------|------------------|-----------------------------------------------------------------------------------------------|---------------------------------------|
| Ammonium Chloride<br>12125-02-9 | -                | LC50: = 725 mg/L, 24h (Lepomis macrochirus)<br>LC50: = 209 mg/L, 96h static (Cyprinus carpio) | LC50: = 202 mg/L, 24h (Daphnia magna) |

### Persistence and Degradability

No information available

### Bioaccumulation/ Accumulation

No information available

### Mobility

No information available.

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste Disposal Methods** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated Packaging** Improper disposal or reuse of this container may be dangerous and illegal.

## SECTION 14: TRANSPORT INFORMATION

**Road and Rail Transport** Not Regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated  
**Marine Pollutant** Not Applicable

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**Legend:** X - Listed '-' - Not Listed

### International Inventories

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

| Component         | CAS-No     | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | AICS |
|-------------------|------------|----------|------|-----------|-------|-----|------|-------|------|------|
| Water             | 7732-18-5  | KE-35400 | X    | 231-791-2 | X     | X   | -    | X     | -    | X    |
| Ammonium Chloride | 12125-02-9 | KE-01645 | X    | 235-186-4 | X     | X   | -    | X     | X    | X    |

| Component         | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water             | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Ammonium Chloride | 12125-02-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component         | CAS-No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|-------------------|------------|----------|------------------------------|---------------------------|
| Water             | 7732-18-5  | Listed   | Not applicable               | Not applicable            |
| Ammonium Chloride | 12125-02-9 | Listed   | Not applicable               | Not applicable            |

## Korean National Regulations

| Component         | CAS-No     | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-------------------|------------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Water             | 7732-18-5  | Annex 1 - KE-35400 Exempt (Index No. 25)                            | Not applicable       | Not applicable                              |
| Ammonium Chloride | 12125-02-9 | Annex 1 - KE-01645                                                  | Not applicable       | Not applicable                              |

| Component         | CAS-No     | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-------------------|------------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Water             | 7732-18-5  | Not applicable                         | Not applicable                              | Not applicable                                  |
| Ammonium Chloride | 12125-02-9 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component         | CAS-No     | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-------------------|------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Water             | 7732-18-5  | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |
| Ammonium Chloride | 12125-02-9 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component         | CAS-No     | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-------------------|------------|-------------------|------------------------------------|-----------------------------------------------------------|
| Water             | 7732-18-5  | Not applicable    | Not applicable                     | Not applicable                                            |
| Ammonium Chloride | 12125-02-9 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component         | CAS-No     | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-------------------|------------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Water             | 7732-18-5  | Not applicable                                               | Not applicable               | Not applicable                         |
| Ammonium Chloride | 12125-02-9 | Not applicable                                               | Not applicable               | Not applicable                         |

| Component         | CAS-No     | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-------------------|------------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Water             | 7732-18-5  | Not applicable                       | Not applicable                                     | Not applicable                     |
| Ammonium Chloride | 12125-02-9 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component         | CAS-No     | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals          | ISHA - Special management materials |
|-------------------|------------|-------------------------------------------------------------|---------------------------------------------------------|-------------------------------------|
| Water             | 7732-18-5  | Not applicable                                              | Not applicable                                          | Not applicable                      |
| Ammonium Chloride | 12125-02-9 | Not applicable                                              | STEL: 20 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> | Not applicable                      |

# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component         | CAS-No     | Class 1 -<br>Oxidising<br>solids | Class 2 -<br>Flammable<br>solid | Class 3 -<br>Spontaneously<br>Combustible<br>Substances<br>and Dangerous<br>Substances<br>When Wet | Class 4 -<br>Flammable<br>liquids | Class 5 -<br>Self-reactive<br>substances | Class 6 -<br>Oxidising<br>liquids |
|-------------------|------------|----------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------|-----------------------------------|
| Water             | 7732-18-5  | Not applicable                   | Not applicable                  | Not applicable                                                                                     | Not applicable                    | Not applicable                           | Not applicable                    |
| Ammonium Chloride | 12125-02-9 | Not applicable                   | Not applicable                  | Not applicable                                                                                     | Not applicable                    | Not applicable                           | Not applicable                    |

## Control Parameters

| Component         | CAS-No     | Korea                                                   | Biological Exposure Indices |
|-------------------|------------|---------------------------------------------------------|-----------------------------|
| Water             | 7732-18-5  | Not listed                                              | Not listed                  |
| Ammonium Chloride | 12125-02-9 | STEL: 20 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> | Not listed                  |

## US Management Information

**OSHA** - Occupational Safety and Health Administration

Not applicable

| Component         | CAS-No     | Specifically Regulated<br>Chemicals | Highly Hazardous Chemicals |
|-------------------|------------|-------------------------------------|----------------------------|
| Water             | 7732-18-5  | Not applicable                      | Not applicable             |
| Ammonium Chloride | 12125-02-9 | Not applicable                      | Not applicable             |

## CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component         | CAS-No     | CERCLA EHS RQs | Hazardous Substances<br>RQs | SARA 313 - Threshold<br>Values % |
|-------------------|------------|----------------|-----------------------------|----------------------------------|
| Water             | 7732-18-5  | Not applicable | Not applicable              | Not applicable                   |
| Ammonium Chloride | 12125-02-9 | Not applicable | 5000 lb                     | 1.0                              |

## CLP Classification - Regulation (EC) No 1272/2008

Based on available data, the classification criteria are not met.

## SECTION 16: OTHER INFORMATION

|                     |                                |
|---------------------|--------------------------------|
| Prepared By         | Regulatory Affairs             |
| Prepared For        | Thermo Fisher Scientific Inc.© |
| Issue Date          | 05-Sep-2019                    |
| Revision Date       | 05-Sep-2019                    |
| Reason for revision | Initial Release.               |

## MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

IMPORTANT: The information contained in this SDS is correct to the best of our knowledge as of the issue date (or subsequent



# SAFETY DATA SHEET

1 ppm Ammonia Standard

Revision Date 05-Sep-2019

---

revision date, if any), and is to be used only as a guide. This SDS does not constitute a guarantee (express or implied) of any kind and we make no warranties of any kind as to the accuracy or completeness of the information contained herein or the merchantability or fitness of the product or this information for a particular purpose. It is the responsibility of each individual buyer/user to determine the suitability of this information and the product for its intended purposes. Product sales are subject to Thermo Fisher Scientifics standard terms and conditions of sale. This information relates only to the designated product as shipped and may not be valid if the product is used in combination with any other materials or is not used in accordance with our instructions, or is altered in any way. It is the responsibility of the buyer/user to ensure that its activities comply with all applicable government requirements. Since conditions of use of the product are not under direct control of Thermo Fisher Scientific, it is the duty of the buyer/user to determine the necessary conditions for the safe use of the product. Thermo Fisher Scientific will not be liable for any injuries or damages resulting from handling, use, misuse or contact with the product.

**End of Safety Data Sheet**